



Title: **Insurance Analytics: Transforming Insurance Data into Actionable Business Insights**

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Introduction

- Our project focuses on Insurance Analytics, which involves analyzing insurance data to gain meaningful insights and support better business decisions.
- The main objective of this project is to understand policy performance, claims patterns, customer behavior, and premium analysis using data analytics techniques.
- We used SQL for data management and querying, Excel for data cleaning and preliminary analysis, and Tableau & Power BI for interactive dashboards and data visualization.
- Through this project, we transformed raw insurance data into visual insights that help in identifying trends, improving risk assessment, and supporting smarter financial decisions in the insurance sector.
- This analytics-driven approach helps organizations enhance efficiency, transparency, and strategic planning in insurance management.

Major KPI's – INSURANCE ANALYTICS

New Placed Achievement %

16.86%

New Invoice Achievement %

4.21%

Cross Sell Placed Achievement %

62.96%

Insurance Business Performance Analytics Dashboard

939

Total Policies

22.65M

Revenue Gap

34

Total Meetings

About KPI's

Insurance Analytics :-

- ✓ It measures the percentage of new policies sold or completed. Here 16.86% means only 16.86% of the target new business has been achieved. Slide Title
- ✓ How much invoicing activity has been completed. Here 4.21% means only a small portion of the invoice target has been reached.
- ✓ Shows how much cross-selling business has been achieved compared to the target. 62.96% means more than half of the cross-selling target has been completed.

Insurance Business Performance Analytics Dashboard :-

- ✓ The total number of insurance policies sold. It counts all policies issued to customers. Here the company has sold 939 policies in total.
- ✓ The difference between the revenue target and the revenue achieved. It represents the amount of revenue still remaining to reach the target. Here 22.65 million revenue is still left to achieve the goal.
- ✓ The total number of meetings held with clients or potential customers. Meetings are important for discussing insurance plans and generating business opportunities. Here 34 meetings have been conducted.

Excel Dashboard



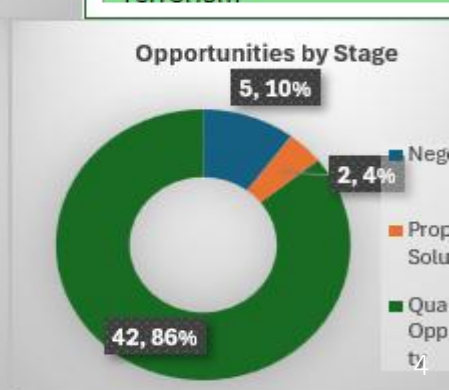
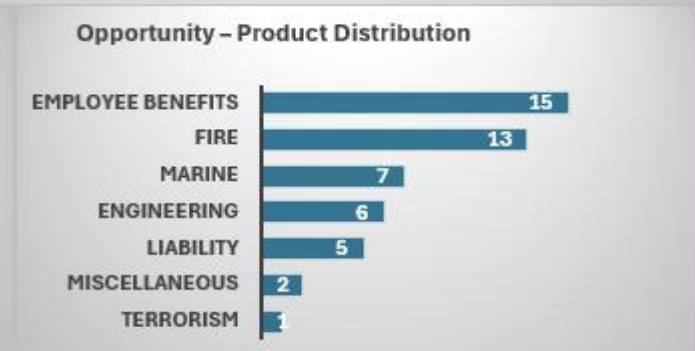
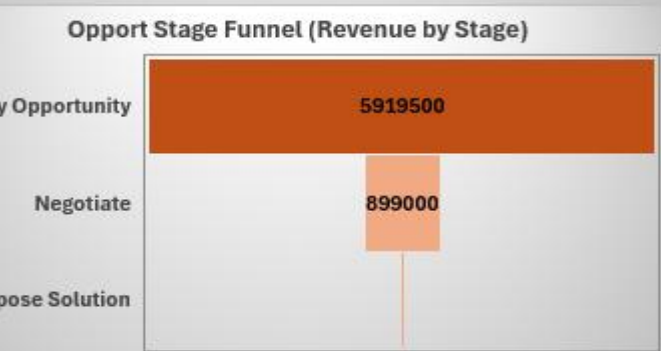
Insurance Dashboard

Refresh

New Placed Achievement %	New Invoice Achievement %	Cross Sell Placed Achievement %	Cross Sell Invoice Achievement %	Renewal Placed Achievement %	Renewal Invoice Achievement %
16.86%	4.21%	62.96%	15.14%	185.87%	45.16%



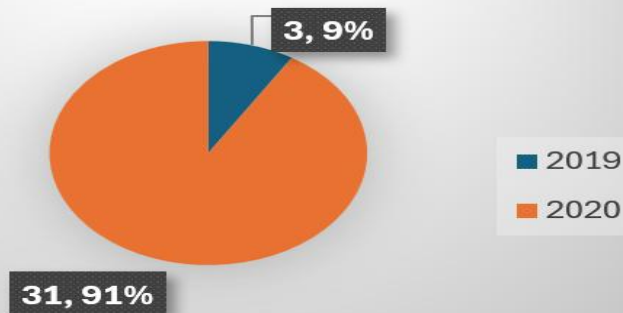
- Business_Type**
- Cross-Sell
 - New
 - Renewal
- product_group**
- Employee Benefits
 - Engineering
 - Fire
 - Liability
 - Marine
 - Miscellaneous
 - Terrorism



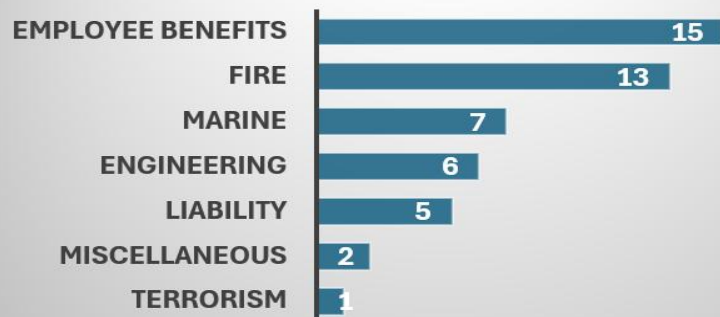
New Business



Meeting Count by Year



Opportunity – Product Distribution



Insights and Suggestions



Insight :-

The achieved value is higher than invoices, but it is still far below the target. There is a large gap between achieved and target.



Suggestion :-

The company should increase sales activities to reach the target. Focus on getting more clients and generating more invoices. The sales team should follow up with potential customers regularly. Monthly progress should be tracked to reduce the gap between achieved and target.



Insight :-

A huge increase in meetings in 2020. More meetings usually mean more opportunities for business growth.



Suggestion :-

Continue the same strategy used in 2020 to keep meeting numbers high. Schedule regular meetings with clients. Encourage the sales team to meet more potential customers. More meetings can help increase sales and achieve the target faster.



Insight :-

Employee Benefits and Fire insurance products are most popular.

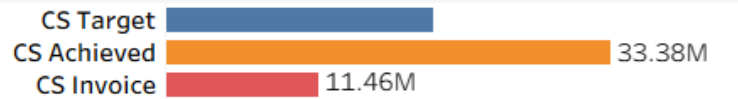


Suggestion :-

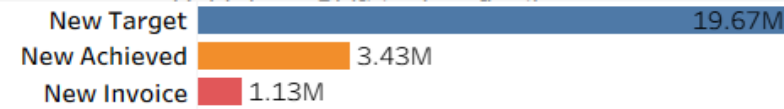
Focus more on Employee Benefits and Fire products because they bring more opportunities. Improve marketing for low-performing products like Miscellaneous and Terrorism.

Tableau dashboard

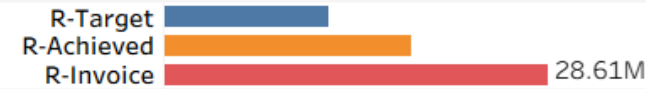
Cross sell



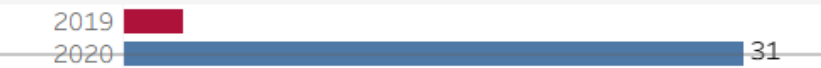
New



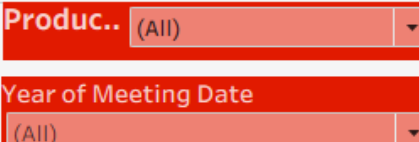
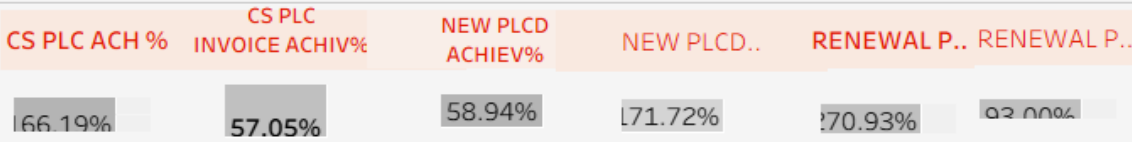
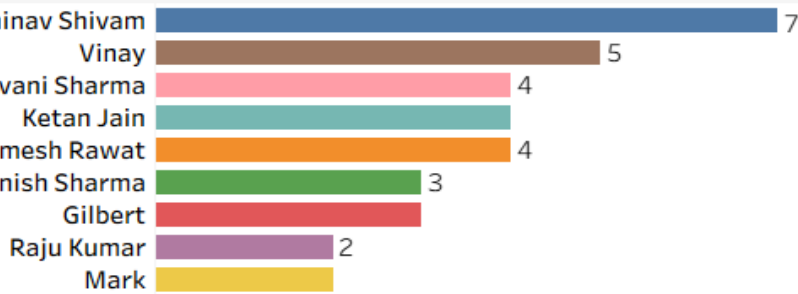
Renewal



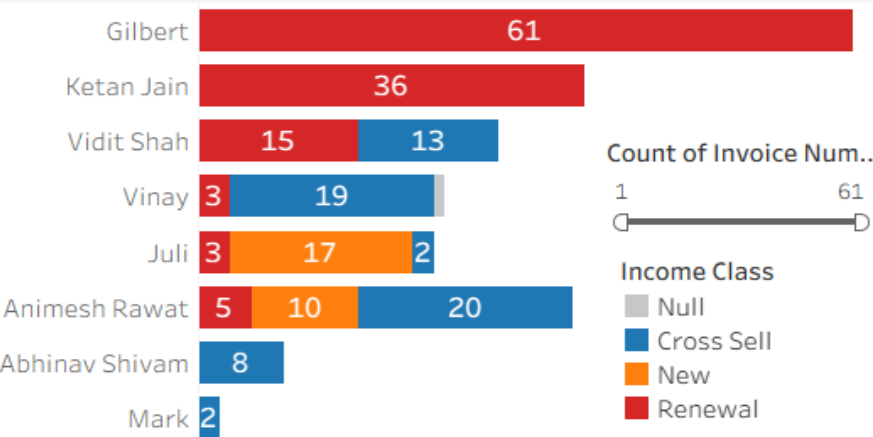
Yearly Meeting Count



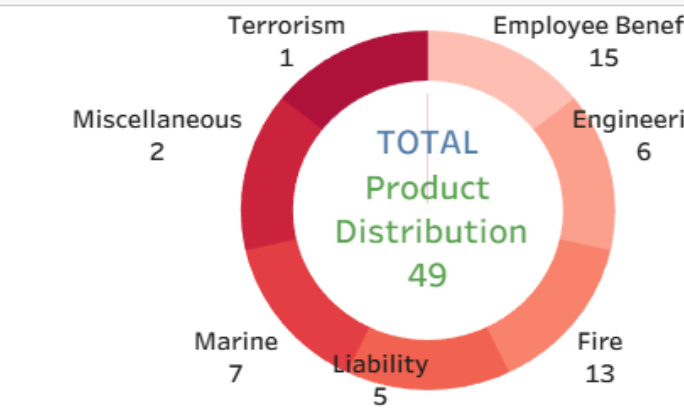
No of Meeting by Accnt Exec



No of Invoice by Account Executive

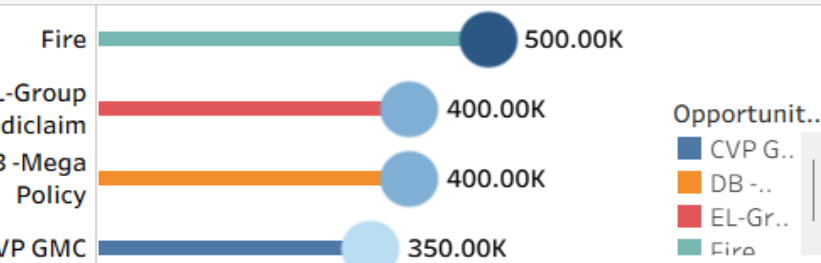


Oppty Product Distribution

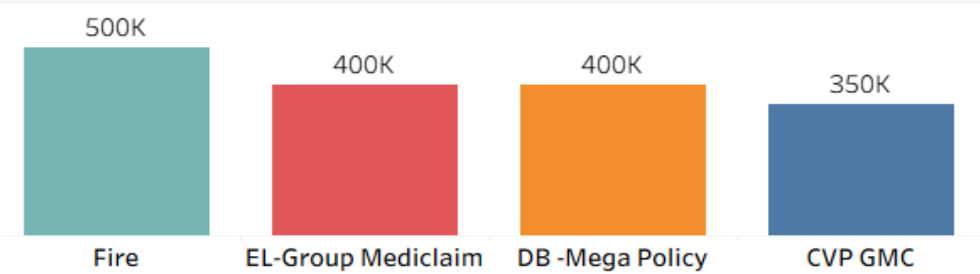


Total Opportu..

Opportunities by Revenue-TOP 4

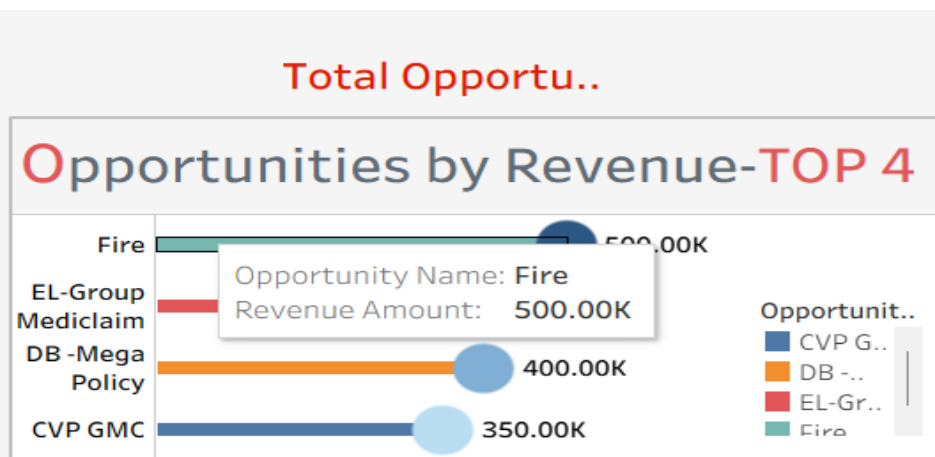
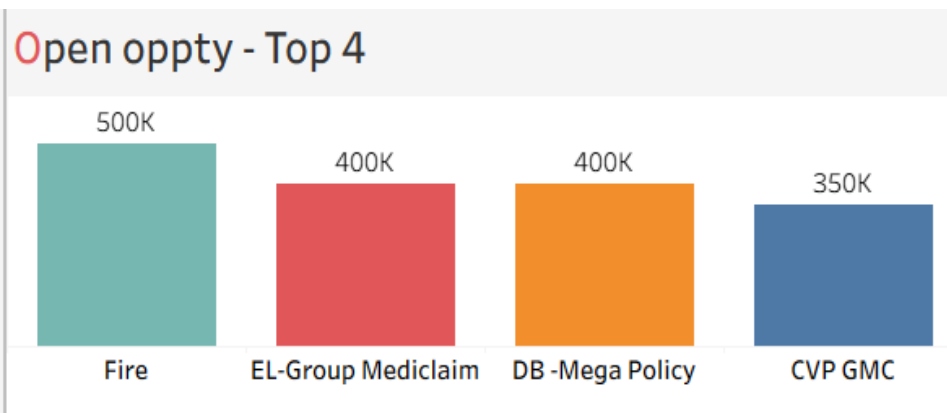
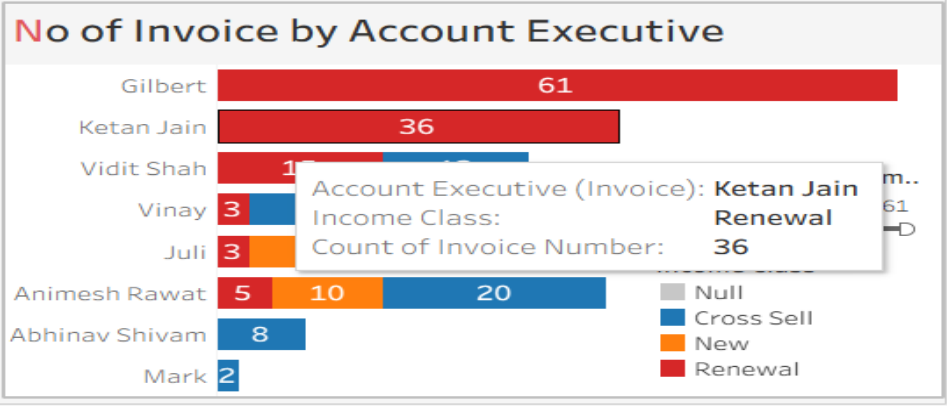


Open oppty - Top 4



Stage Funnle by Revnue





Insights and Suggestions

Insight:-

Gilbert has the highest number of invoices (61), followed by Ketan Jain (36). Other executives have much lower invoice counts.

Suggestion:-

Top performers can share their strategies with the team. Other executives should increase client meetings and follow-ups to generate more invoices.

Insight:-

Fire has the highest open opportunity (500K), followed by EL-Group Mediclaim and DB-Mega Policy (400K each).

Suggestion:-

The company should focus more on Fire insurance opportunities. Sales teams should prioritize these high-value opportunities to increase revenue.

Insight:-

Fire generates the highest revenue opportunity (500K), while CVP GMC has the lowest among the top four (350K).

Suggestion:-

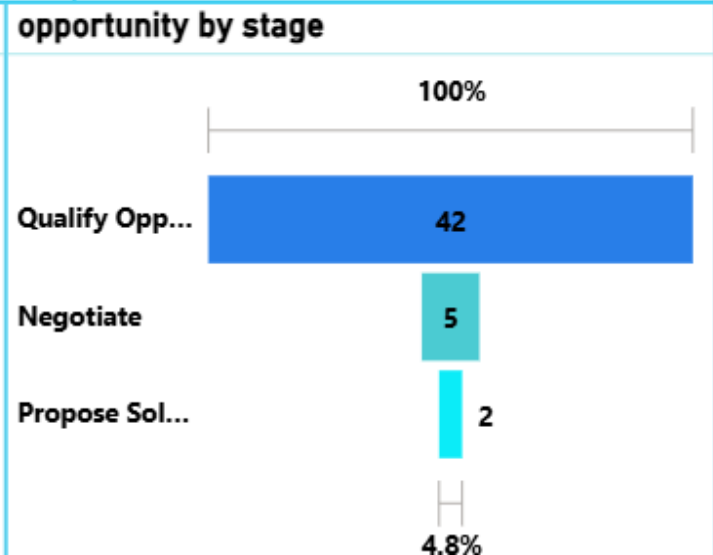
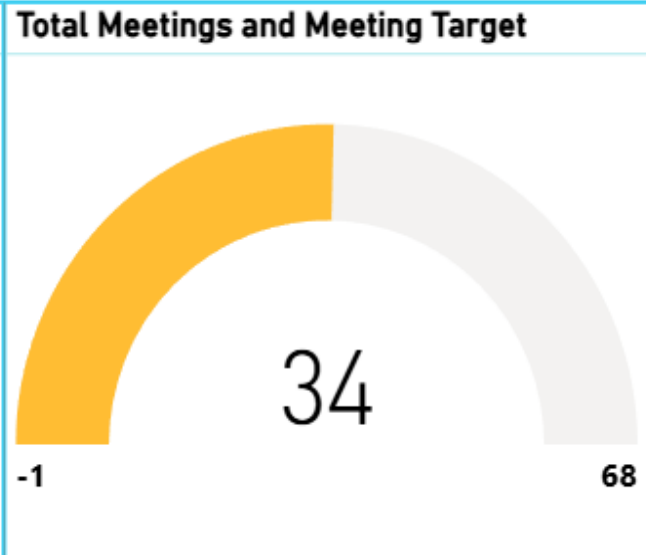
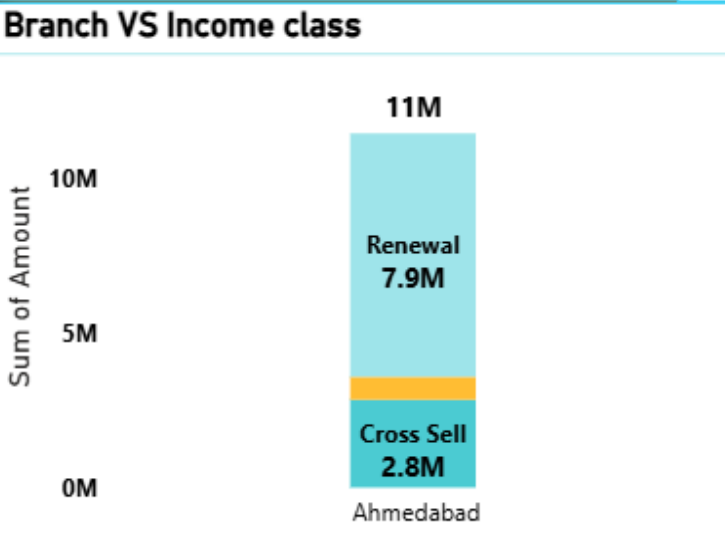
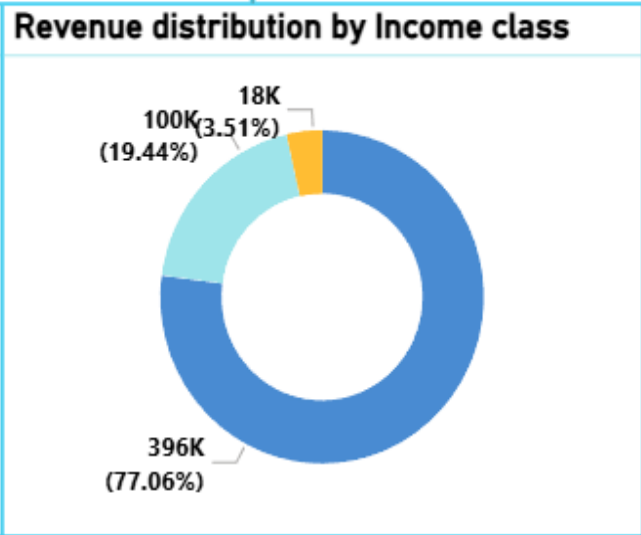
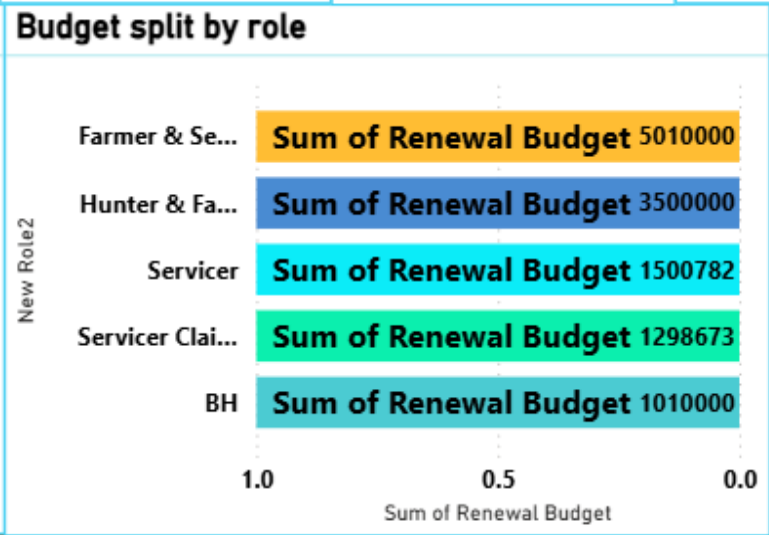
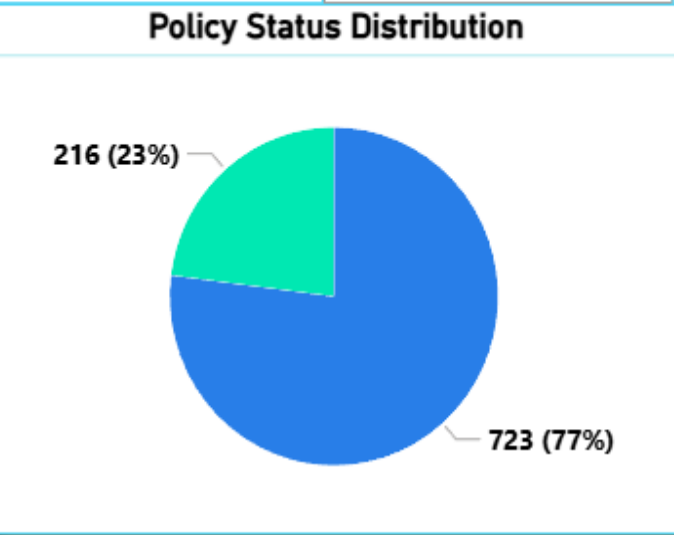
Sales teams should focus on converting Fire and EL-Group Mediclaim opportunities. Proper follow-ups can help turn these opportunities into actual sales.

POWER BI DASHBOARD

Insurance Business Performance Analytics Dashboard



939	515K	12M	22.65M	34	49
Total Policies	Revenue	Total Renewal bud...	Revenue Gap	Total Meetings	Total Opportunities



Filter by Account Executive

Exe Name

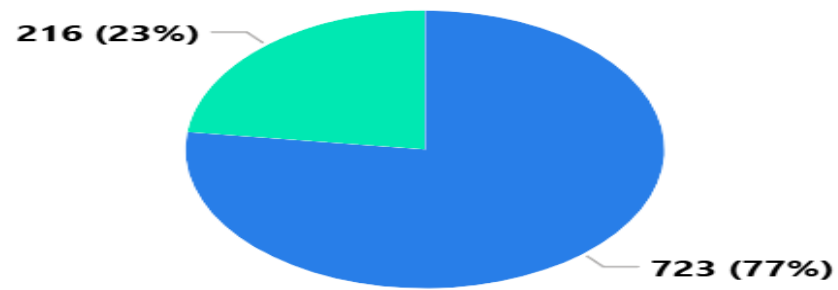
- ☐ Abhinav Shivam
- ☐ Animesh Rawat
- ☐ Gilbert
- ☐ Juli
- ☐ Ketan Jain
- ☐ Kumar Jha
- ☐ Manish Sharma
- ☐ Mark
- ☐ Raju Kumar
- ☐ Shivani Sharma
- ☐ Vididt Saha
- ☐ Vinay

Select Income class

income_class

- ☐ (Blank)
- ☐ Cross Sell
- ☐ New
- ☐ Renewal

Policy Status Distribution



Insights and Suggestions

Insight:-

77% of policies are active (723) and 23% are inactive (216).



Suggestion:-

Target the inactive 23% with renewal campaigns to shift them to active status.



Insight:-

Ahmedabad branch earns 11M total, split into 7.9M renewals and 2.8M cross-sell.



Suggestion:-

Strengthen renewal processes and increase cross-sell efforts to raise the 11M revenue further.



Insight:-

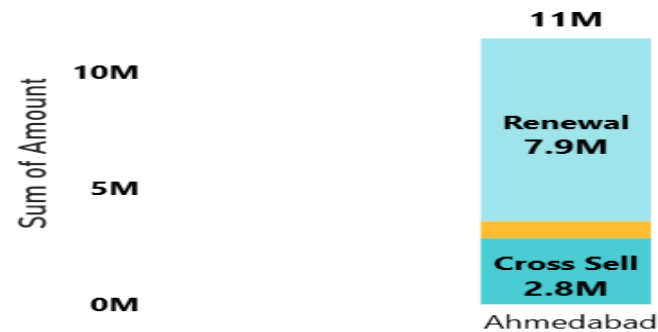
77.06% of revenue comes from 396K high-income customers, 19.44% from 100K mid-income, and 3.51% from 18K low-income.



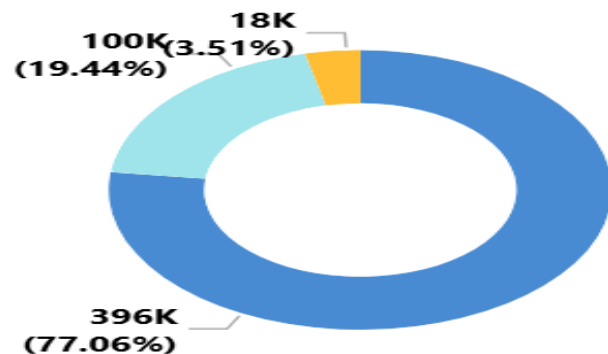
Suggestion:-

Focus marketing on high-income segments while nurturing mid-income growth to balance revenue.

Branch VS Income class



Revenue distribution by Income class



SQL QUERIES

UNION ALL, LEFT JOIN, SUM(), GROUP BY

```
1 • DESCRIBE opportunity;
2 • DESCRIBE invoice;
3 • DESCRIBE fees;
4 • DESCRIBE brokerage;
5 • DESCRIBE `individual budgets`;
6
7 ##1.Create Combined Revenue Subquery
8
9 • SELECT
10     `Account Exe ID`,
11     revenue_transaction_type,
12     Amount
13 FROM invoice
14
15 UNION ALL
16
17 SELECT
18     `Account Exe ID`,
19     revenue_transaction_type,
20     Amount
21 FROM fees
22
```

```
SELECT
    `Account Exe ID`,
    revenue_transaction_type,
    Amount
FROM brokerage;

##2.Now Aggregate Revenue Type Wise

SELECT
    `Account Exe ID`,

    SUM(CASE
        WHEN revenue_transaction_type = 'New'
        THEN Amount ELSE 0 END) AS New_Achieved,

    SUM(CASE
        WHEN revenue_transaction_type = 'Cross Sell'
        THEN Amount ELSE 0 END) AS Cross_Achieved,

    SUM(CASE
        WHEN revenue_transaction_type = 'Renewal'
        THEN Amount ELSE 0 END) AS Renewal_Achieved
FROM (
```

```
    COALESCE(r.New_Achieved,0) AS New_Achieved,
    COALESCE(r.Cross_Achieved,0) AS Cross_Achieved,
    COALESCE(r.Renewal_Achieved,0) AS Renewal_Achieved

FROM `individual budgets` ib

LEFT JOIN (

    SELECT
        `Account Exe ID`,

        SUM(CASE WHEN revenue_transaction_type = 'New'
            THEN Amount ELSE 0 END) AS New_Achieved,

        SUM(CASE WHEN revenue_transaction_type = 'Cross Sell'
            THEN Amount ELSE 0 END) AS Cross_Achieved,

        SUM(CASE WHEN revenue_transaction_type = 'Renewal'
            THEN Amount ELSE 0 END) AS Renewal_Achieved

    FROM (
        SELECT `Account Exe ID`, revenue_transaction_type, Amount FROM invoice
        UNION ALL
```

SUMMARY

- ✓ This project focuses on Insurance Analytics, where we analyzed insurance data to gain meaningful insights into policy performance, claims, and customer trends.
- ✓ We used Excel for data cleaning and initial analysis, SQL for managing and querying the insurance database, and Tableau and Power BI to create interactive dashboards and visualizations.
- ✓ Through this analysis, we examined key metrics such as total premium, claim amounts, policy distribution, and customer behavior to understand the overall performance of the insurance business. The dashboards help transform raw insurance data into clear visual insights, enabling better risk assessment, financial analysis, and data-driven decision making.
- ✓ Overall, this project demonstrates how data analytics tools can improve business understanding and support strategic decisions in the insurance industry.

CONCLUSION

- ✓ In this project, we analyzed insurance data to understand policy performance, premium collection, and claim patterns.
- ✓ By using tools like Excel, SQL, Tableau, and Power BI, we converted raw data into meaningful insights and interactive dashboards.
- ✓ The analysis helped us identify important metrics such as total premium, claim amount, and policy distribution, which are useful for evaluating the performance of the insurance business.



THANK YOU